

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5
Marks:25

Total

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Santosh(192521007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

3) Assume a real-time OS used in embedded system controlling industrial machinery.

a) Type of Operating System

→ Real-Time Operating System (RTOS)

Reason:

- * Fast Response
- * Predictable execution.
- * Meets deadlines
- * Used in embedded and industrial systems

b) Diagram

User Process

kernel

Scheduler

Memory Manager

Device Driver

Hardware

c) Synchronization System Calls

- * fork()
- * wait()
- * sem_wait()
- * sem_post()
- * pthread_mutex_lock()
- * pthread_mutex_unlock()

These synchronize multiple process and prevent race conditions.

4) A distributed computing system consists of multiple interconnected computers sharing resources over a network.

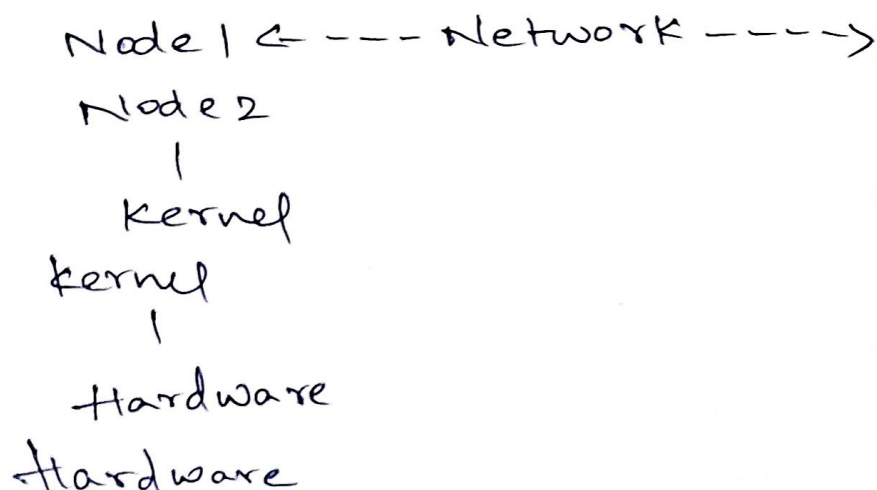
a) Type of Operating System:

Distributed Operating System

Comparison

Distributed OS	Centralized OS
Multiple computers	Single computer
Shares resources	Local resources
High reliability	Less reliable
Faster resource sharing	Limited sharing

b) Distributed OS Architecture



c) IPC System call Sequence

pipe()

fork()

write()

read()

close()

or

socket()

bind()

listen()

accept()

send()

recv()

close()